ZACHARY RODRIGUEZ, PhD

zachary.rodriguez@pennmedicine.upenn.edu · zachrodriguez.me · github.com/frogriguez · Philadelphia, PA

SUMMARY

Senior Bioinformatician with 10+ years of end-to-end research experience spanning large-scale clinical biobank informatics, comparative genomics, phylogenomics, and computational mental health. Demonstrated breadth across biological domains with a consistent throughline: designing and deploying scalable pipelines, applying ML and LLMs to complex biological questions, and building tools that make computational biology accessible to experimentalists and clinicians alike. Currently leading agentic AI development for undiagnosed phenotyping across 300K+ biobank participants.

PROFESSIONAL EXPERIENCE

Senior Bioinformatician | Penn Medicine Biobank, University of Pennsylvania | Philadelphia, PA **2023–present**

- Designed and lead the PMBB Geno-Pheno Toolkit — a suite of production-grade Nextflow DSL2 pipelines (GWAS, ExWAS, PheWAS, gene burden, PRS, VEP, meta-analysis) deployed across local HPC, AWS, GCP, Azure, DNAnexus, and the All of Us Research Workbench; benchmarked on 300K+ participants and 164M+ variants.
- Leading one of PMBB's largest agentic AI initiatives: applying LLM-based sequential diagnosis frameworks (inspired by Nori et al., 2025) to scan unstructured EHR notes and clinical records across 300K+ participants, identifying patients with undiagnosed phenotypes through iterative, evidence-driven hypothesis generation.
- Curates, processes, and QCs the annual multi-modal PMBB data release (phenotype, exome, genotype, imputed, clinical EHR) for 300K+ biobank participants; manages genomic data requests, cohort queries, and variant-level extractions for the research community.
- Oversees statistical analysis and bioinformatics workflows for 5+ cross-functional research teams; enforces QA/QC standards, SOP documentation, and reproducibility across GWAS, ExWAS, and multi-omic association studies.
- Develops tools extending bioinformatics accessibility to non-computational researchers — NEAT-Plots (open-source Python visualization library), containerized Docker/Singularity pipelines, interactive dashboards — while mentoring 6 researchers and guest lecturing at Perelman School of Medicine (BMIN 5020, 2025).
- Leads monthly Scientific Brain Trust Workshops advancing genomics literacy, QC practices, and responsible AI use across the PMBB research community.

Postdoctoral Researcher in Computational Psychopathology | Center for Computation & Technology, LSU | Baton Rouge, LA **2020–2023**

- Built ML and NLP pipelines for clinical digital phenotyping from speech, video, and EHR data; co-developed a multimodal smartphone app for passive mental health monitoring; resulted in 7 peer-reviewed publications in Schizophrenia Bulletin, Schizophrenia Research, and Psychiatry Research.
- Supervised statistical frameworks and ML models across 5 concurrent clinical research projects under HIPAA/IRB-compliant governance; maintained documented, version-controlled codebases across multi-institutional collaborations.

NSF Graduate Research Fellow | Museum of Natural Science, Louisiana State University | Baton Rouge, LA **2014–2020**

- Applied NGS, whole-genome sequencing, de novo genome assembly (SPAdes, Trinity, Canu), phylogenomics, and comparative genomics to investigate convergent evolution of adaptive traits in reptiles across the tree of life; first-author publication in Science Advances (NPR, The Atlantic, AP coverage).
- Secured $180,000+ in competitive funding including NSF GRFP ($102K); managed 280,000 CPU-hour HPC allocations across 3 independent projects; mentored 9 trainees from high school through graduate level.

TECHNICAL SKILLS

Programming & Tools: Python, R, bash/shell, SQL, HTML/CSS, Nextflow (DSL2), Snakemake, Docker, Singularity, Git, LLMs, agentic AI frameworks, AWS, GCP, Azure, DNAnexus

Bioinformatics & Genomics: NGS, GWAS, ExWAS, PheWAS, SAIGE, PLINK 2.0, GATK, BWA, samtools, BLAST, VEP, genome assembly, variant annotation, phylogenomics, comparative genomics

ML & Clinical Informatics: NLP, ML (supervised/unsupervised), LLM application & evaluation, EHR integration, HIPAA/IRB compliance, ETL, clinical digital phenotyping

Communication: Scientific writing, SOP authorship, data visualization (ggplot2, NEAT-Plots), mentoring, public science communication

SELECTED PUBLICATIONS (12 PEER-REVIEWED; 2 UNDER REVIEW)

- Palmer, D. S. et al. (2026). The Biobank Rare Variant consortium powers the discovery of rare genetic associations through global collaboration. medRxiv. medRxiv. doi: 10.64898/2026.05.21.26353759.
- Verma, A., Rodriguez, Z.B., Guare, L., et al. (2024). Command line to pipeLine: Cross-biobank analyses with Nextflow. PSB 2025 Proceedings, 696–701.
- Cohen, A.S., Rodriguez, Z.B., et al. (2022). Natural Language Processing and Psychosis: On the Need for Comprehensive Psychometric Evaluation', Schizophrenia bulletin. doi: 10.1093/schbul/sbac051.
- Rodriguez, Z. (2020) Evolution of Green Blood in New Guinea Lizards: Phylogenomics, Biogeography, and Comparative Genomics. Louisiana State University and Agricultural & Mechanical College. Thesis.

Full list: scholar.google.com/citations?user=R0HlvQIAAAAJ | github.com/PMBB-Informatics-and-Genomics

SELECTED PRESENTATIONS & WORKSHOPS

- Rodriguez, Z.B., Verma, A., et al. (2025). Command Line to PipeLine: Cross-biobank analyses with Nextflow. Workshop led at PSB 2025.
- Rodriguez, Z.B., et al. (2024). Pipeline Palooza: Streamlining multi-omic discoveries across biobanks. ASHG 2024. [Poster, first author]
- Rodriguez, Z.B., et al. (2024). Bearing the Load: Allostatic Load and Health Outcomes in Biobank-Linked EHR. Dept. of Medicine Research Day, UPenn. [Poster, first author]

EDUCATION

PhD, Biology | Louisiana State University | Baton Rouge, LA **2020**

NSF Graduate Research Fellow · Dissertation: Evolution of Green Blood in New Guinea Lizards (Phylogenomics, Biogeography, Comparative Genomics)

BS, Biological Sciences | Clarkson University | Potsdam, NY **2013**